

Dathan Lang

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EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

B.S. in Computer Science and ITWS GPA: 3.87/4.00

Expected May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture & Operating Systems, Web Systems Development, Discrete Mathematics, and Applied Quantum Computing

EXPERIENCE

Software Engineering Intern

May 2025 – Aug. 2025

Opex Corporation

Moorestown, NJ

- Developed a high-performance C++ encoder manager for the EAGLE system, harnessing AES and Blowfish encryption algorithms to process over 16,000 log entries per hour.
- Orchestrated pipeline optimization techniques in the Falcon+ system to resolve job synchronization errors and maximize memory management, eliminating bottlenecks.
- Overhauled CertainScan GUI to enhance functionality by integrating personalized paper coloring settings and streamlining the interface for real-time operator print jobs.

Backend Web Developer

Apr 2025 – Aug 2025

Presentation Assistants

Troy, NY

- Spearheaded a scalable PHP/SQL learning platform in an Agile environment, restructuring code to accelerate database queries and bolster system security during high-demand sessions.
- Implemented an LLM pipeline using RAG techniques to generate teaching lessons automatically during the lecture that dynamically adapts to student questions.

Data Science Researcher

Oct 2023 – May 2024

University of Pennsylvania

Philadelphia, PA

- Constructed a multi-threaded Java engine simulating over 10,000 games of Scrabble to collect metrics from millions of moves to inform game theory research.
- Automated a stochastic ETL process to analyze how 3-letter word strategies optimize board control and leaves, producing a 5,000-word research report.

PROJECTS

TrailMate | *HTML/CSS, JavaScript, Node.js, MapBox, Supabase*

Feb 28 – Mar 01, 2026

- Won HenHack's **Best Use of Snowflake AI** for an accessibility-focused AI-driven campus walking platform.
- Leveraged Snowflake Cortex to analyze live routes to identify optimal navigation and popular walking paths.
- Deployed Augmented Reality and ElevenLabs to provide personalized multi-language and audiovisual guidance based on user profiles.

ByteBites | *HTML/CSS, JavaScript, Node.js, React, SQL*

Nov 15 – 16, 2025

- Won HackRPI's **Best Data Science Hack** for a dynamic student meal prepping and nutrition app.
- Synthesized an automated extraction tool to transform dining hall data into structured meal plans.
- Formulated a modular user interface that will automate personalization based on updated individual health restrictions and physical objectives.

TECHNICAL SKILLS

Languages: C/C++, Java, Python, SQL (Postgres), JavaScript, TypeScript, HTML/CSS, PHP, R

Frameworks: React, Node.js, Next.js, JUnit, Express.js, FastAPI, Flutter

Developer Tools: Git, Docker, Plastic SCM, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Libraries: pandas, NumPy, Matplotlib, scikit-learn, boost

AWARDS & EXTRACURRICULAR ACTIVITIES

Awards: HenHack Best Use of Snowflake AI (2026), HackRPI Best Data Science Hack (2025), Dean's Honor List (2024–Present), Rensselaer Medalist (2024), 2x U.S. Candidate at World Youth Scrabble Championship (2022-2023)

Extracurriculars: CSCI 1200 Data Structures Mentoring TA, Google Developer Groups, 8th Street Jazz Band (Lead Pianist), Rensselaer Music Association (Security Coordinator)